Bayesian Optimization Report: Function 2

GP Health Summary per Iteration

Iteration	GP Health	Status
1	0.746	Good
2	0.746	Good
3	0.746	Good
4	0.746	Good
5	0.746	Good
6	0.746	Good
7	0.746	Good
8	0.746	Good
9	0.746	Good
10	0.746	Good
11	0.746	Good
12	0.746	Good
13	0.746	Good
14	0.746	Good
15	0.748	Good
16	0.745	Good
17	0.746	Good
18	0.744	Good
19	0.743	Good
20	0.742	Good
21	0.741	Good
22	0.741	Good
23	0.741	Good
24	0.740	Good
25	0.739	Good
26	0.738	Good
27	0.737	Good
28	0.736	Good
29	0.734	Good
30	0.734	Good

GP Sigma Summary

Iteration	Mean Sigma	Min Sigma	Max Sigma
1	0.2484	0.2251	0.2547
2	0.2328	0.1949	0.2447
3	0.2245	0.1912	0.2360
4	0.2112	0.1686	0.2279
5	0.1927	0.1474	0.2207
6	0.1820	0.1329	0.2142
7	0.1763	0.1286	0.2087
8	0.1625	0.1206	0.2033
9	0.1523	0.1097	0.1972
10	0.1457	0.1117	0.1858
11	0.1399	0.1076	0.1814
12	0.1273	0.0926	0.1706
13	0.1171	0.0859	0.1622
14	0.1054	0.0815	0.1400
15	0.0997	0.0883	0.1234
16	0.0958	0.0754	0.1310
17	0.0932	0.0803	0.1159
18	0.0900	0.0749	0.1146
19	0.0866	0.0708	0.1134
20	0.0830	0.0685	0.1105
21	0.0799	0.0657	0.1086
22	0.0769	0.0645	0.1045
23	0.0740	0.0633	0.0984
24	0.0706	0.0597	0.0971
25	0.0693	0.0577	0.0963
26	0.0657	0.0541	0.0947
27	0.0647	0.0516	0.0924
28	0.0613	0.0495	0.0905
29	0.0614	0.0480	0.0904
30	0.0585	0.0461	0.0900

Best Results Per Iteration

Iteration	Best Input	Best Output	Kernel	Acquisition	GP Health	Anomalies
1	[0.24848914 0.74394629]	0.28434552319503065	RBF	UCB	0.746	{'type': 'noise_collapse', 'severity': 0.8, 'details': {'noise_level': 1e-08}}
2	[0.77717878 0.67854085]	0.29011589388732334	RBF	UCB	0.746	{'type': 'noise_collapse', 'severity': 0.8, 'details': {'noise_level': 1e-08}}
3	[0.07385101 0.51287097]	0.2810242008466332	RBF	UCB	0.746	{'type': 'noise_collapse', 'severity': 0.8, 'details': {'noise_level': 1e-08}}
4	[0.50871514 0.01087674]	0.2783529596553811	RBF	UCB	0.746	{'type': 'noise_collapse', 'severity': 0.8, 'details': {'noise_level': 1e-08}}
5	[0.19776516 0.04715067]	0.2766864164054726	RBF	UCB	0.746	{'type': 'noise_collapse', 'severity': 0.8, 'details': {'noise_level': 1e-08}}
6	[0.62573043 0.71972519]	0.2860737593571882	RBF	UCB	0.746	{'type': 'noise_collapse', 'severity': 0.8, 'details': {'noise_level': 1e-08}}
7	[0.39168011 0.23723834]	0.2816087716264945	RBF	UCB	0.746	{'type': 'noise_collapse', 'severity': 0.8, 'details': {'noise_level': 1e-08}}
8	[0.94604214 0.62959721]	0.26286415019174914	RBF	UCB	0.746	{'type': 'noise_collapse', 'severity': 0.8, 'details': {'noise_level': 1e-08}}
9	[0.28839886 0.05056745]	0.2706378918163851	RBF	UCB	0.746	{'type': 'noise_collapse', 'severity': 0.8, 'details': {'noise_level': 1e-08}}
10	[0.81160673 0.62381616]	0.2638896097655656	RBF	UCB	0.746	{'type': 'noise_collapse', 'severity': 0.8, 'details': {'noise_level': 1e-08}}
11	[0.10534758 0.38731663]	0.2282240213322314	RBF	UCB	0.746	{'type': 'noise_collapse', 'severity': 0.8, 'details': {'noise_level': 1e-08}}
12	[0.74918889 0.68978693]	0.3381617372771361	RBF	UCB	0.746	{'type': 'noise_collapse', 'severity': 0.8, 'details': {'noise_level': 1e-08}}

13	[0.04528136 0.29673389]	0.21042394656453167	RBF	UCB	0.746	{'type': 'noise_collapse', 'severity': 0.8, 'details': {'noise_level': 1e-08}}
14	[0.22260792 0.72273811]	0.28752430521571365	RBF	UCB	0.746	{'type': 'noise_collapse', 'severity': 0.8, 'details': {'noise_level': 1e-08}}
15	[0.6598388 0.00552307]	0.49586665021163834	RBF	UCB	0.748	{'type': 'noise_collapse', 'severity': 0.8, 'details': {'noise_level': 1e-08}}
16	[0.48439776 0.35077021]	0.2575664310484821	RBF	UCB	0.745	{'type': 'noise_collapse', 'severity': 0.8, 'details': {'noise_level': 1e-08}}
17	[0.64303733 0.32445959]	0.37113709276019985	RBF	UCB	0.746	{'type': 'noise_collapse', 'severity': 0.8, 'details': {'noise_level': 1e-08}}
18	[0.86317153 0.23933983]	0.39324687533675035	RBF	UCB	0.744	{'type': 'noise_collapse', 'severity': 0.8, 'details': {'noise_level': 1e-08}}
19	[0.41411447 0.51638578]	0.28940124492683683	RBF	UCB	0.743	{'type': 'noise_collapse', 'severity': 0.8, 'details': {'noise_level': 1e-08}}
20	[0.27113003 0.10997108]	0.29687122436315455	RBF	UCB	0.742	{'type': 'noise_collapse', 'severity': 0.8, 'details': {'noise_level': 1e-08}}
21	[0.96321556 0.64058834]	0.21786814994080808	RBF	UCB	0.741	{'type': 'noise_collapse', 'severity': 0.8, 'details': {'noise_level': 1e-08}}
22	[0.5266677 0.341291]	0.21426557661714254	RBF	UCB	0.741	{'type': 'noise_collapse', 'severity': 0.8, 'details': {'noise_level': 1e-08}}
23	[0.7894659 0.71953667]	0.2875536351659622	RBF	UCB	0.741	{'type': 'noise_collapse', 'severity': 0.8, 'details': {'noise_level': 1e-08}}
24	[0.76043778 0.60197787]	0.31088407049919425	RBF	UCB	0.740	{'type': 'noise_collapse', 'severity': 0.8, 'details': {'noise_level': 1e-08}}
25	[0.0856632 0.02134384]	0.28825048297406947	RBF	UCB	0.739	{'type': 'noise_collapse', 'severity': 0.8, 'details': {'noise_level': 1e-08}}

26	[0.21191465 0.41669923]	0.29407251778034527	RBF	UCB	0.738	{'type': 'noise_collapse', 'severity': 0.8, 'details': {'noise_level': 1e-08}}
27	[0.84370058 0.0074451]	0.3006578318997816	RBF	UCB	0.737	{'type': 'noise_collapse', 'severity': 0.8, 'details': {'noise_level': 1e-08}}
28	[0.73193147 0.74725731]	0.4005650611617765	RBF	UCB	0.736	{'type': 'noise_collapse', 'severity': 0.8, 'details': {'noise_level': 1e-08}}
29	[0.62527002 0.15905035]	0.28526298942342493	RBF	UCB	0.734	{'type': 'noise_collapse', 'severity': 0.8, 'details': {'noise_level': 1e-08}}
30	[0.39389161 0.45289621]	0.2853416074706266	RBF	UCB	0.734	{'type': 'noise_collapse', 'severity': 0.8, 'details': {'noise_level': 1e-08}}

Acquisition Values Per Iteration

Iteration 1

Method	Max Value
UCB	0.808382
EI	0.003117
PI	0.031143
THOMPSON	0.649467
KG	0.003117
ENTROPY	1.472225
PORTFOLIO	1.015247

Iteration 2

Method	Max Value
UCB	0.450952
EI	0.002499
PI	0.026568
THOMPSON	0.747695
KG	0.002499
ENTROPY	1.317408

PORTFOLIO	0.871603

Iteration 3

Method	Max Value
UCB	0.422608
EI	0.001668
PI	0.018695
THOMPSON	0.640987
KG	0.001668
ENTROPY	1.356358
PORTFOLIO	0.883774

Iteration 4

Method	Max Value
UCB	0.421445
EI	0.001317
PI	0.015735
THOMPSON	0.564881
KG	0.001317
ENTROPY	1.323812
PORTFOLIO	0.782448

Iteration 5

Method	Max Value
UCB	0.417764
EI	0.001067
PI	0.013586
THOMPSON	0.531596
KG	0.001067
ENTROPY	1.230266
PORTFOLIO	0.697693

Iteration 6

Method	Max Value
UCB	0.414563

EI	0.000879
PI	0.011876
THOMPSON	0.726485
KG	0.000879
ENTROPY	1.200280
PORTFOLIO	0.656331

Iteration 7

Method	Max Value
UCB	0.410442
EI	0.000716
PI	0.009868
THOMPSON	0.704343
KG	0.000716
ENTROPY	1.192541
PORTFOLIO	0.614127

Iteration 8

Method	Max Value
UCB	0.406538
EI	0.000494
PI	0.007284
THOMPSON	0.525717
KG	0.000494
ENTROPY	1.073984
PORTFOLIO	0.566198

Iteration 9

Method	Max Value
UCB	0.403667
EI	0.000404
PI	0.006211
THOMPSON	0.505542
KG	0.000404
ENTROPY	1.150892

PORTFOLIO	0.534887

Iteration 10

Method	Max Value
UCB	0.400156
EI	0.000219
PI	0.004267
THOMPSON	0.530075
KG	0.000219
ENTROPY	1.061976
PORTFOLIO	0.465261

Iteration 11

Method	Max Value
UCB	0.395038
EI	0.000156
PI	0.003252
THOMPSON	0.514699
KG	0.000156
ENTROPY	1.009851
PORTFOLIO	0.448677

Iteration 12

Method	Max Value
UCB	0.389877
EI	0.000109
PI	0.002409
THOMPSON	0.518753
KG	0.000109
ENTROPY	1.006948
PORTFOLIO	0.420457

Iteration 13

Method	Max Value
UCB	0.395754

EI	0.000015
PI	0.000370
THOMPSON	0.452537
KG	0.000015
ENTROPY	0.932595
PORTFOLIO	0.398565

Iteration 14

Method	Max Value
UCB	0.412300
EI	0.000005
PI	0.000168
THOMPSON	0.468632
KG	0.000005
ENTROPY	0.922317
PORTFOLIO	0.349698

Iteration 15

Method	Max Value
UCB	0.450938
EI	0.000053
PI	0.001995
THOMPSON	0.478449
KG	0.000053
ENTROPY	0.955862
PORTFOLIO	0.399312

Iteration 16

Method	Max Value
UCB	0.383756
EI	0.000001
PI	0.000025
THOMPSON	0.377310
KG	0.000001
ENTROPY	0.866184

PORTFOLIO	0.301989

Iteration 17

Method	Max Value
UCB	0.352797
EI	0.000000
PI	0.000002
THOMPSON	0.367850
KG	0.000000
ENTROPY	0.862189
PORTFOLIO	0.306849

Iteration 18

Method	Max Value
UCB	0.360499
EI	0.000000
PI	0.000001
THOMPSON	0.356599
KG	0.000000
ENTROPY	0.851145
PORTFOLIO	0.299597

Iteration 19

Method	Max Value
UCB	0.299001
EI	0.000000
PI	0.000000
THOMPSON	0.308288
KG	0.000000
ENTROPY	0.860731
PORTFOLIO	0.257502

Iteration 20

Method	Max Value
UCB	0.291527

EI	0.000000
PI	0.000000
THOMPSON	0.350109
KG	0.000000
ENTROPY	0.838604
PORTFOLIO	0.236101

Iteration 21

Method	Max Value
UCB	0.275848
EI	0.000000
PI	0.000000
THOMPSON	0.330398
KG	0.000000
ENTROPY	0.807479
PORTFOLIO	0.221698

Iteration 22

Method	Max Value
UCB	0.275824
EI	0.000000
PI	0.000000
THOMPSON	0.347151
KG	0.000000
ENTROPY	0.812100
PORTFOLIO	0.227284

Iteration 23

Method	Max Value
UCB	0.273806
EI	0.000000
PI	0.000000
THOMPSON	0.320969
KG	0.000000
ENTROPY	0.796404

PORTFOLIO	0.230524

Iteration 24

Method	Max Value
UCB	0.283111
EI	0.000000
PI	0.000000
THOMPSON	0.291162
KG	0.000000
ENTROPY	0.785538
PORTFOLIO	0.215984

Iteration 25

Method	Max Value
UCB	0.266953
EI	0.000000
PI	0.000000
THOMPSON	0.287142
KG	0.000000
ENTROPY	0.785239
PORTFOLIO	0.217917

Iteration 26

Method	Max Value
UCB	0.265078
EI	0.000000
PI	0.000000
THOMPSON	0.299566
KG	0.000000
ENTROPY	0.784576
PORTFOLIO	0.196380

Iteration 27

Method	Max Value
UCB	0.267669

EI	0.000000
PI	0.000000
THOMPSON	0.322642
KG	0.000000
ENTROPY	0.778304
PORTFOLIO	0.196371

Iteration 28

Method	Max Value
UCB	0.338806
EI	0.000000
PI	0.000000
THOMPSON	0.314054
KG	0.000000
ENTROPY	0.787236
PORTFOLIO	0.240136

Iteration 29

Method	Max Value
UCB	0.252437
EI	0.000000
PI	0.000000
THOMPSON	0.307938
KG	0.000000
ENTROPY	0.768648
PORTFOLIO	0.191363

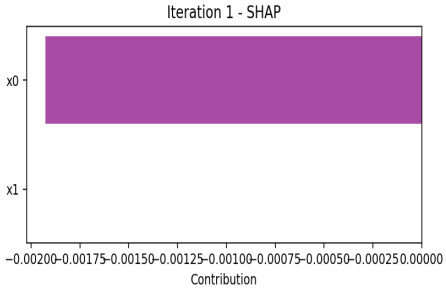
Iteration 30

Method	Max Value
UCB	0.250078
EI	0.000000
PI	0.000000
THOMPSON	0.281078
KG	0.000000
ENTROPY	0.763182

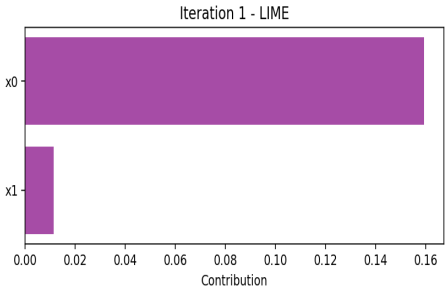
PORTFOLIO	0.178540

Feature Explanations (SHAP / LIME)

Iteration 1 - SHAP



Iteration 1 - LIME



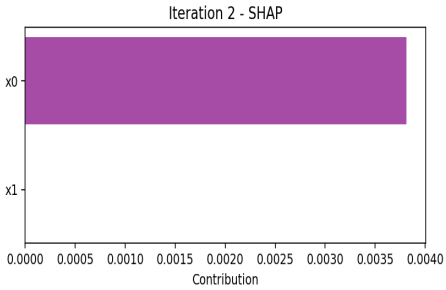
Iteration 1 - SHAP Values

Feature	SHAP Value
x0	-0.00193
x1	0.00000

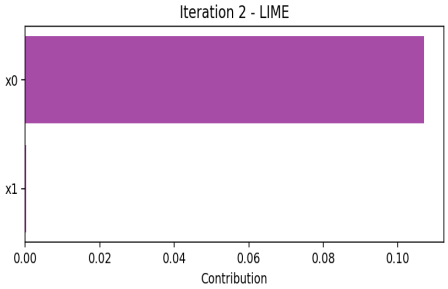
Iteration 1 - LIME Values

Feature	LIME Value
x0	0.15929
x1	0.01143

Iteration 2 - SHAP



Iteration 2 - LIME



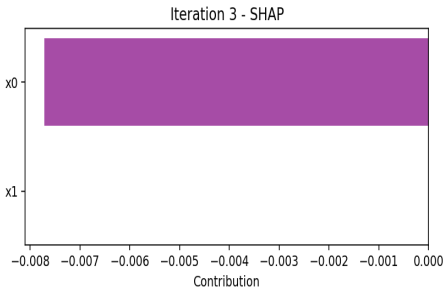
Iteration 2 - SHAP Values

Feature	SHAP Value
x0	0.00381
x1	0.00000

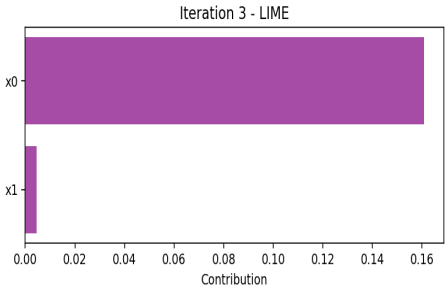
Iteration 2 - LIME Values

Feature	LIME Value
x0	0.10701
x1	0.00028

Iteration 3 - SHAP



Iteration 3 - LIME



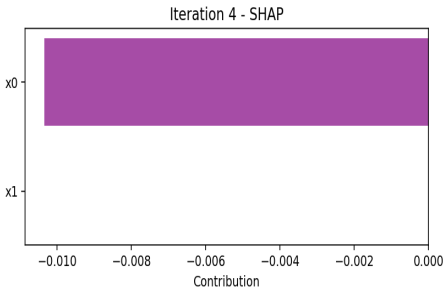
Iteration 3 - SHAP Values

Feature	SHAP Value
x0	-0.00771
x1	0.00000

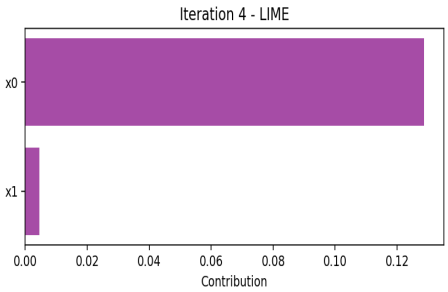
Iteration 3 - LIME Values

Feature	LIME Value
x0	0.16075
x1	0.00465

Iteration 4 - SHAP



Iteration 4 - LIME



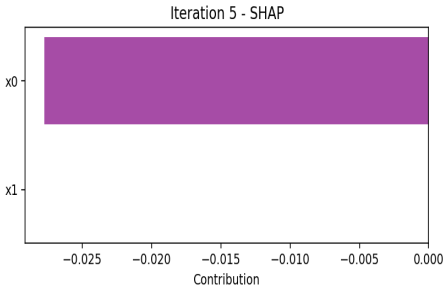
Iteration 4 - SHAP Values

Feature	SHAP Value
x0	-0.01034
x1	0.00000

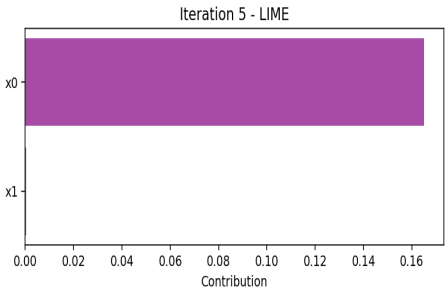
Iteration 4 - LIME Values

Feature	LIME Value
x0	0.12865
x1	0.00460

Iteration 5 - SHAP



Iteration 5 - LIME



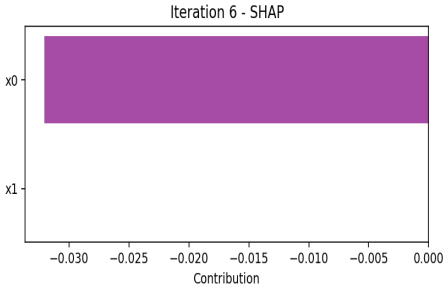
Iteration 5 - SHAP Values

Feature	SHAP Value
x0	-0.02773
x1	0.00000

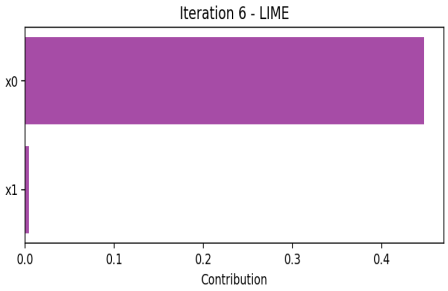
Iteration 5 - LIME Values

Feature	LIME Value
x0	0.16500
x1	0.00027

Iteration 6 - SHAP



Iteration 6 - LIME



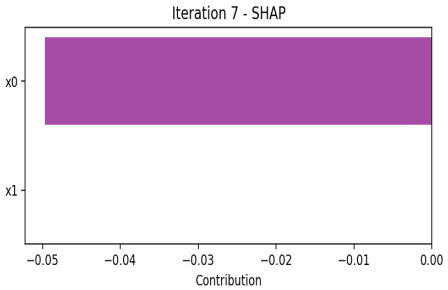
Iteration 6 - SHAP Values

Feature	SHAP Value
x0	-0.03206
x1	0.00000

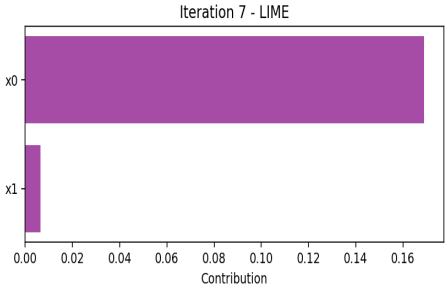
Iteration 6 - LIME Values

Feature	LIME Value
x0	0.44734
x1	0.00409

Iteration 7 - SHAP



Iteration 7 - LIME



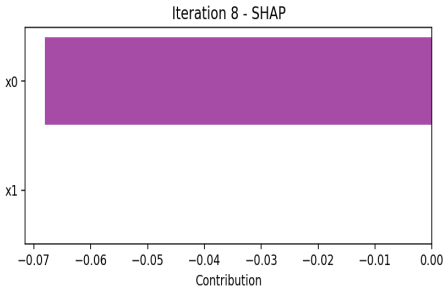
Iteration 7 - SHAP Values

Feature	SHAP Value
x0	-0.04971
x1	0.00000

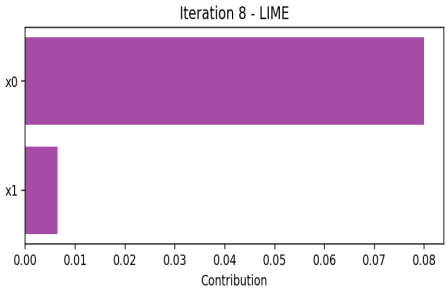
Iteration 7 - LIME Values

Feature	LIME Value
x0	0.16889
x1	0.00638

Iteration 8 - SHAP



Iteration 8 - LIME



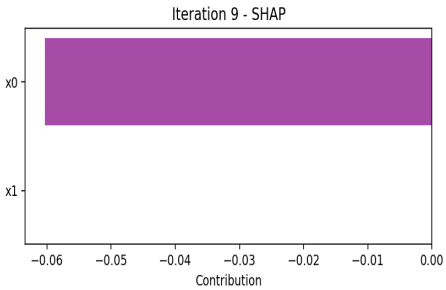
Iteration 8 - SHAP Values

Feature	SHAP Value
x0	-0.06810
x1	0.00000

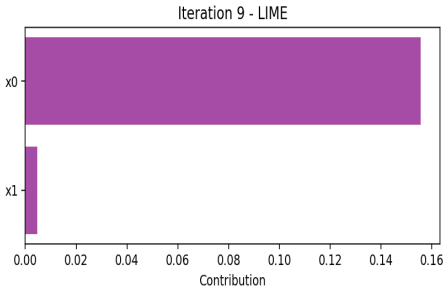
Iteration 8 - LIME Values

Feature	LIME Value
x0	0.07993
x1	0.00643

Iteration 9 - SHAP



Iteration 9 - LIME



Iteration 9 - SHAP Values

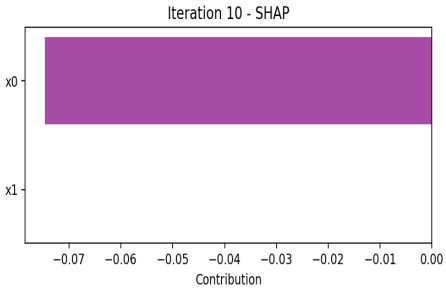
Feature	SHAP Value
x0	-0.06036
x1	0.00000

Iteration 9 - LIME Values

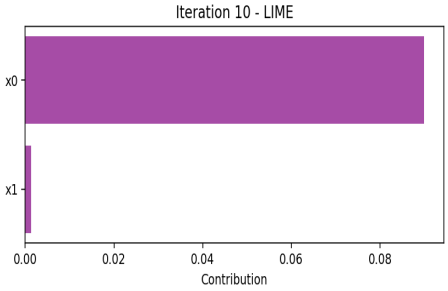
Feature	LIME Value
x0	0.15554

x1	0.00459

Iteration 10 - SHAP



Iteration 10 - LIME



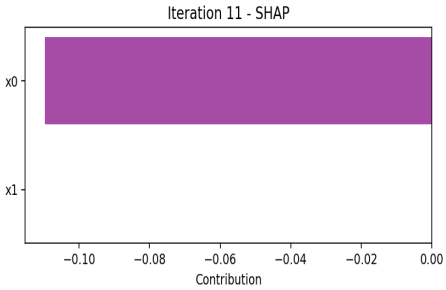
Iteration 10 - SHAP Values

Feature	SHAP Value
x0	-0.07468
x1	0.00000

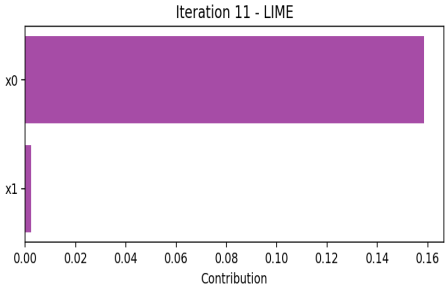
Iteration 10 - LIME Values

Feature	LIME Value
x0	0.08991
x1	0.00134

Iteration 11 - SHAP



Iteration 11 - LIME



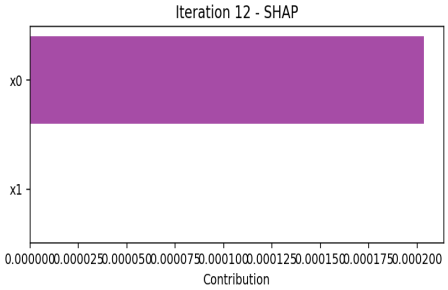
Iteration 11 - SHAP Values

Feature	SHAP Value
x0	-0.10976
x1	0.00000

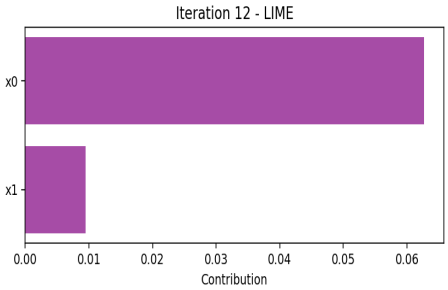
Iteration 11 - LIME Values

Feature	LIME Value
x0	0.15848
x1	0.00227

Iteration 12 - SHAP



Iteration 12 - LIME



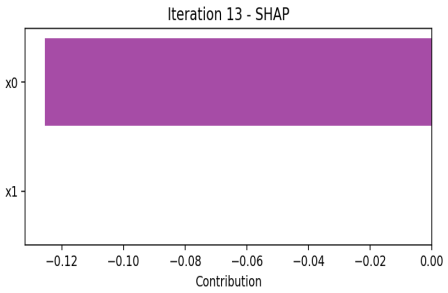
Iteration 12 - SHAP Values

Feature	SHAP Value
x0	0.00020
x1	0.00000

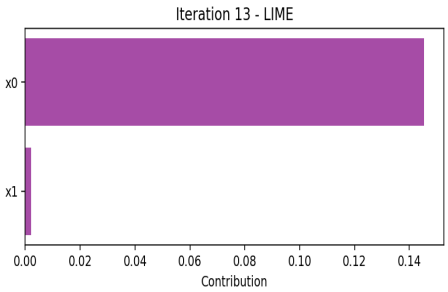
Iteration 12 - LIME Values

Feature	LIME Value
x0	0.06266
x1	0.00951

Iteration 13 - SHAP



Iteration 13 - LIME



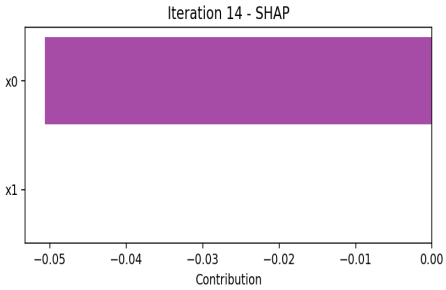
Iteration 13 - SHAP Values

Feature	SHAP Value
x0	-0.12561
x1	0.00000

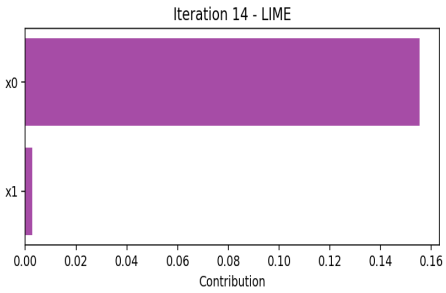
Iteration 13 - LIME Values

Feature	LIME Value
x0	0.14528
x1	0.00205

Iteration 14 - SHAP



Iteration 14 - LIME



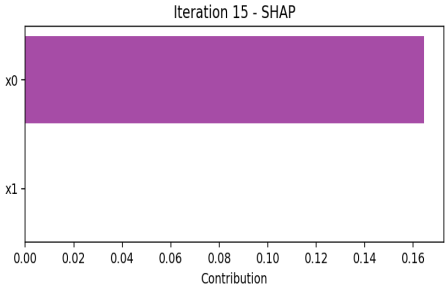
Iteration 14 - SHAP Values

Feature	SHAP Value
x0	-0.05066
x1	0.00000

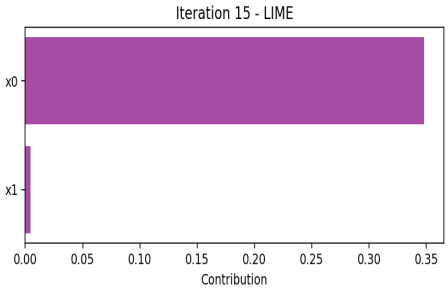
Iteration 14 - LIME Values

Feature	LIME Value
x0	0.15530
x1	0.00261

Iteration 15 - SHAP



Iteration 15 - LIME



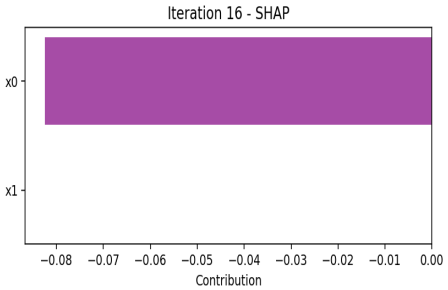
Iteration 15 - SHAP Values

Feature	SHAP Value
x0	0.16440
x1	0.00000

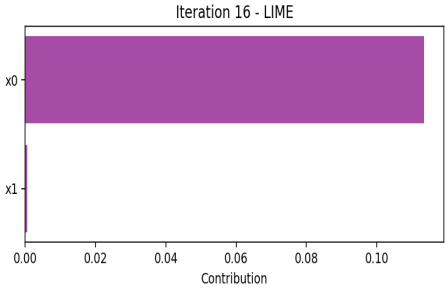
Iteration 15 - LIME Values

Feature	LIME Value
x0	0.34796
x1	0.00445

Iteration 16 - SHAP



Iteration 16 - LIME



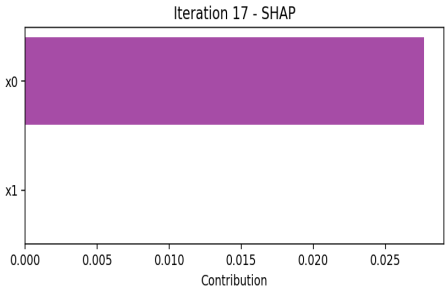
Iteration 16 - SHAP Values

Feature	SHAP Value
x0	-0.08251
x1	0.00000

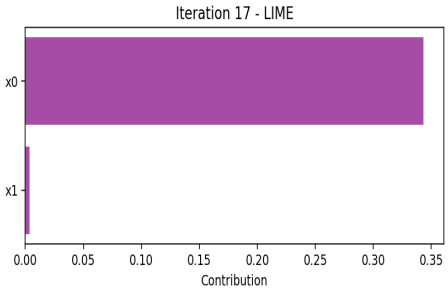
Iteration 16 - LIME Values

Feature	LIME Value
x0	0.11345
x1	0.00047

Iteration 17 - SHAP



Iteration 17 - LIME



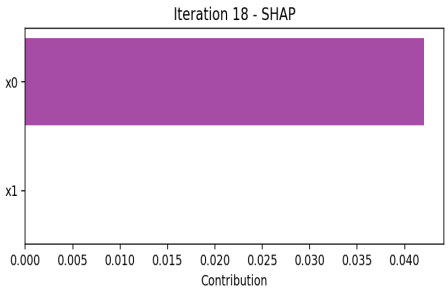
Iteration 17 - SHAP Values

Feature	SHAP Value
x0	0.02767
x1	0.00000

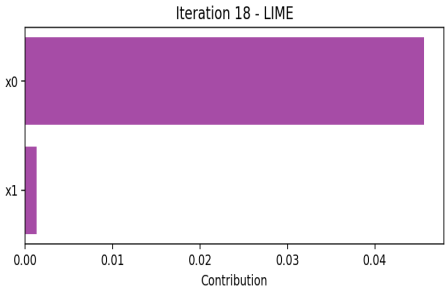
Iteration 17 - LIME Values

Feature	LIME Value
x0	0.34339
x1	0.00337

Iteration 18 - SHAP



Iteration 18 - LIME



Iteration 18 - SHAP Values

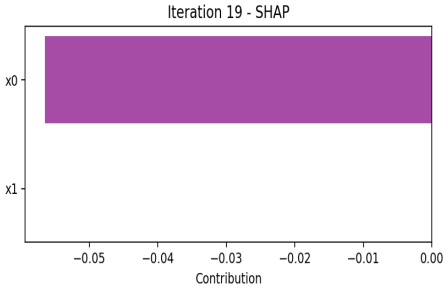
Feature	SHAP Value
x0	0.04203
x1	0.00000

Iteration 18 - LIME Values

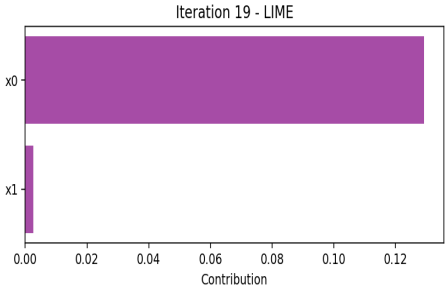
Feature	LIME Value
x0	0.04561

x1	0.00126

Iteration 19 - SHAP



Iteration 19 - LIME



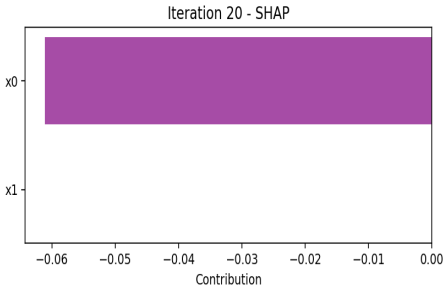
Iteration 19 - SHAP Values

Feature	SHAP Value
x0	-0.05653
x1	0.00000

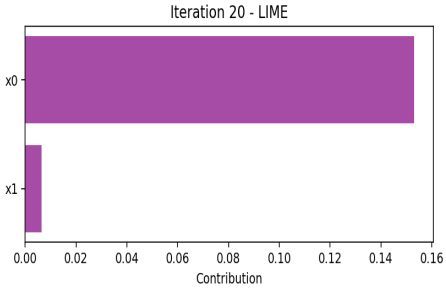
Iteration 19 - LIME Values

Feature	LIME Value
x0	0.12919
x1	0.00260

Iteration 20 - SHAP



Iteration 20 - LIME



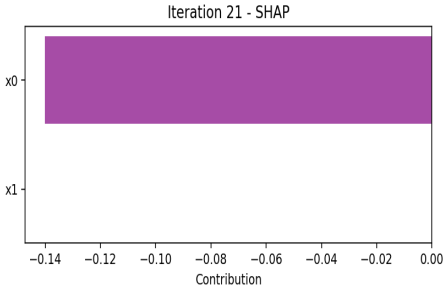
Iteration 20 - SHAP Values

Feature	SHAP Value
x0	-0.06117
x1	0.00000

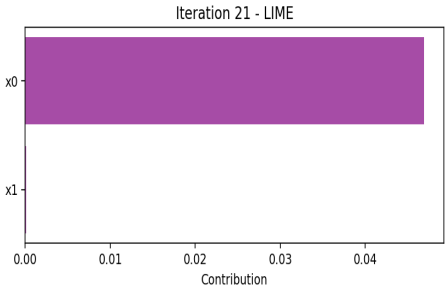
Iteration 20 - LIME Values

Feature	LIME Value
x0	0.15306
x1	0.00649

Iteration 21 - SHAP



Iteration 21 - LIME



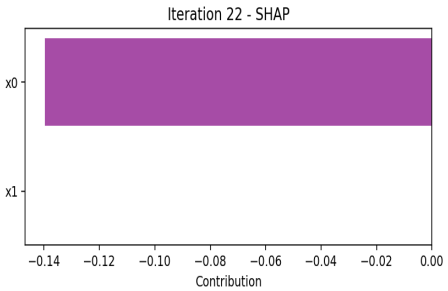
Iteration 21 - SHAP Values

Feature	SHAP Value
x0	-0.14017
x1	0.00000

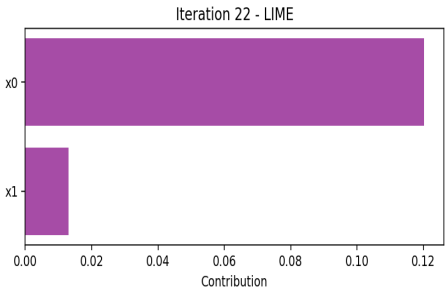
Iteration 21 - LIME Values

Feature	LIME Value
x0	0.04688
x1	0.00012

Iteration 22 - SHAP



Iteration 22 - LIME



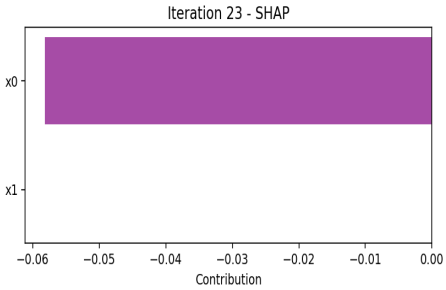
Iteration 22 - SHAP Values

Feature	SHAP Value
x0	-0.13975
x1	0.00000

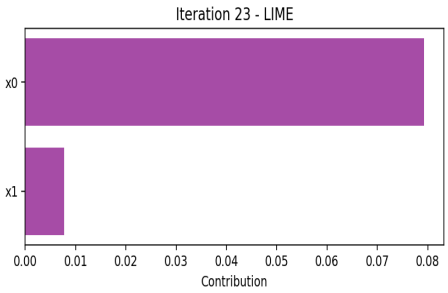
Iteration 22 - LIME Values

Feature	LIME Value
x0	0.12014
x1	0.01301

Iteration 23 - SHAP



Iteration 23 - LIME



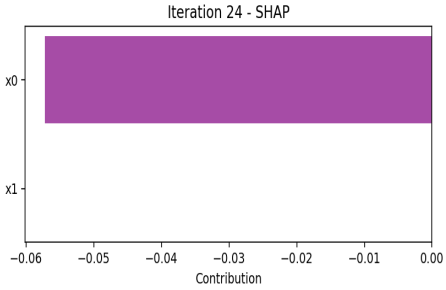
Iteration 23 - SHAP Values

Feature	SHAP Value
x0	-0.05818
x1	0.00000

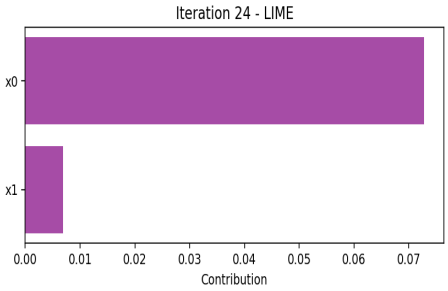
Iteration 23 - LIME Values

Feature	LIME Value
x0	0.07925
x1	0.00772

Iteration 24 - SHAP



Iteration 24 - LIME



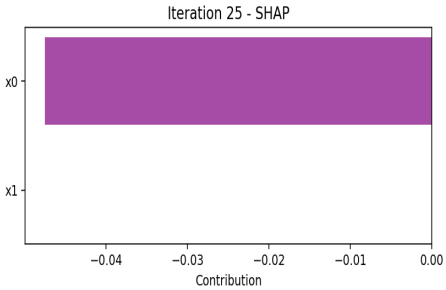
Iteration 24 - SHAP Values

Feature	SHAP Value
x0	-0.05727
x1	0.00000

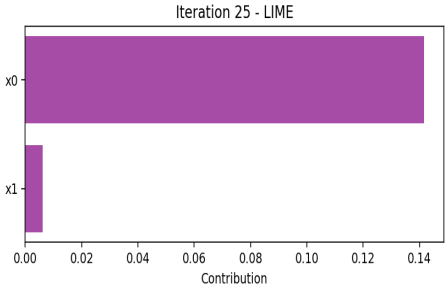
Iteration 24 - LIME Values

Feature	LIME Value
x0	0.07275
x1	0.00693

Iteration 25 - SHAP



Iteration 25 - LIME



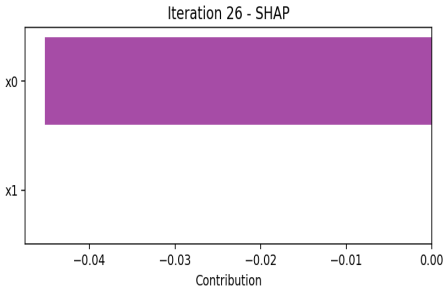
Iteration 25 - SHAP Values

Feature	SHAP Value
x0	-0.04756
x1	0.00000

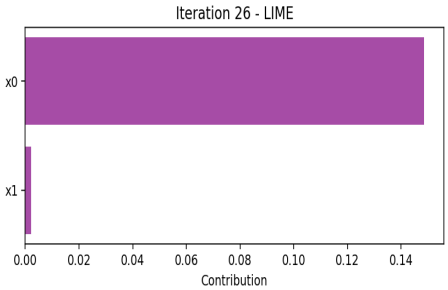
Iteration 25 - LIME Values

Feature	LIME Value
x0	0.14154
x1	0.00618

Iteration 26 - SHAP



Iteration 26 - LIME



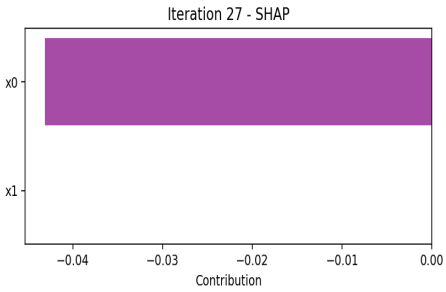
Iteration 26 - SHAP Values

Feature	SHAP Value
x0	-0.04523
x1	0.00000

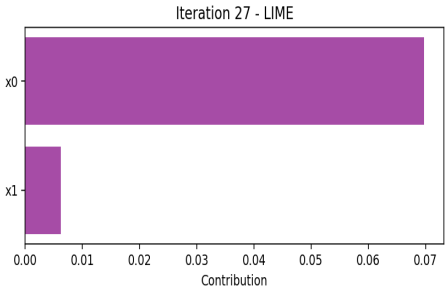
Iteration 26 - LIME Values

Feature	LIME Value
x0	0.14846
x1	0.00213

Iteration 27 - SHAP



Iteration 27 - LIME



Iteration 27 - SHAP Values

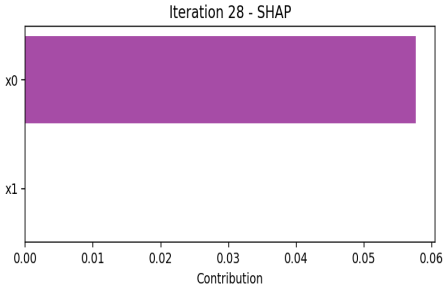
Feature	SHAP Value
x0	-0.04314
x1	0.00000

Iteration 27 - LIME Values

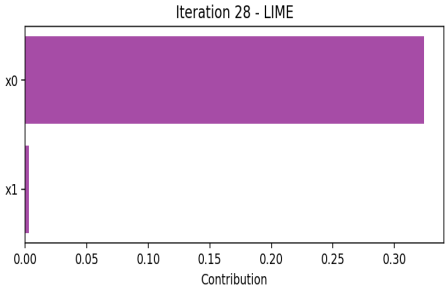
Feature	LIME Value
x0	0.06973

x1	0.00622

Iteration 28 - SHAP



Iteration 28 - LIME



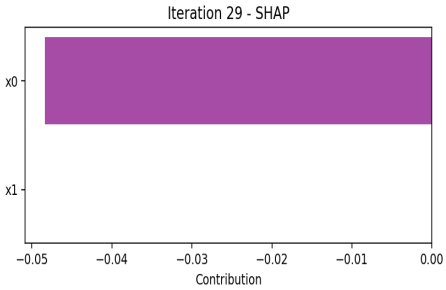
Iteration 28 - SHAP Values

Feature	SHAP Value
x0	0.05766
x1	0.00000

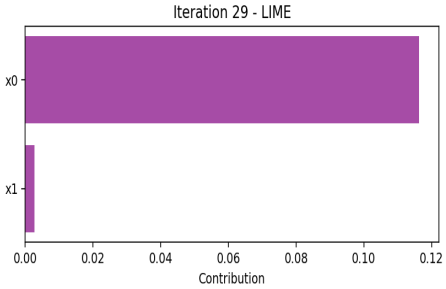
Iteration 28 - LIME Values

Feature	LIME Value
x0	0.32393
x1	0.00289

Iteration 29 - SHAP



Iteration 29 - LIME



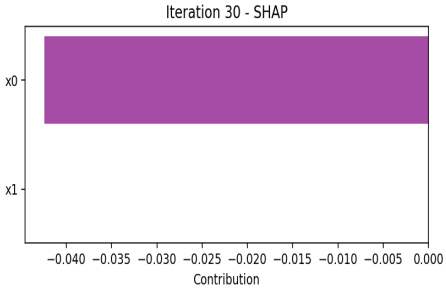
Iteration 29 - SHAP Values

Feature	SHAP Value
x0	-0.04840
x1	0.00000

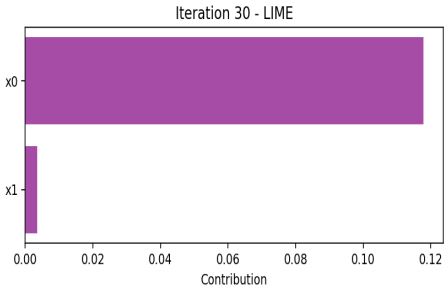
Iteration 29 - LIME Values

Feature	LIME Value
x0	0.11639
x1	0.00276

Iteration 30 - SHAP



Iteration 30 - LIME



Iteration 30 - SHAP Values

Feature	SHAP Value
x0	-0.04240
x1	0.00000

Iteration 30 - LIME Values

Feature	LIME Value
x0	0.11785
x1	0.00348